

- 2 A metal block, of 15 cm height, was partially submerged in water of density $1.0 \times 10^3 \text{ kg m}^{-3}$. The block has a uniform density, ρ , and a uniform cross-sectional area, A . The block is held up by an inelastic cord tied to a mass hanger, through two frictionless pulleys, as shown in Fig. 2.1.

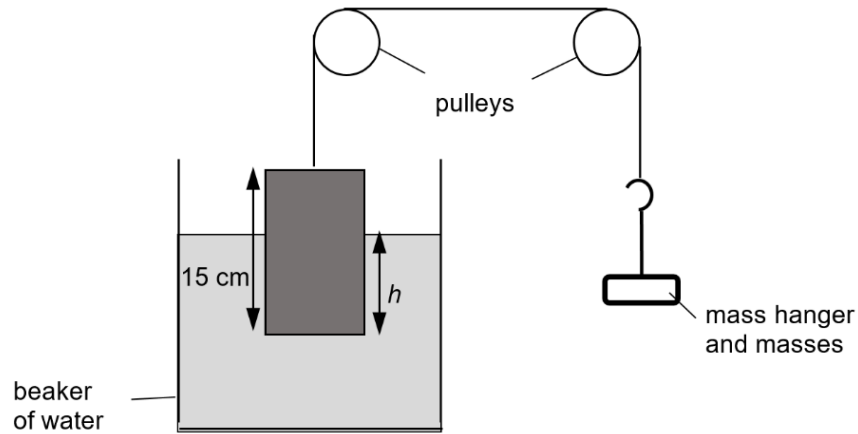


Fig. 2.1

Different quantities of masses were added to the mass hanger and the corresponding submerged height, h , of the block was measured.

The variation of total mass m of mass hanger and masses added, with the submerged height, h , is shown in Fig. 2.2.

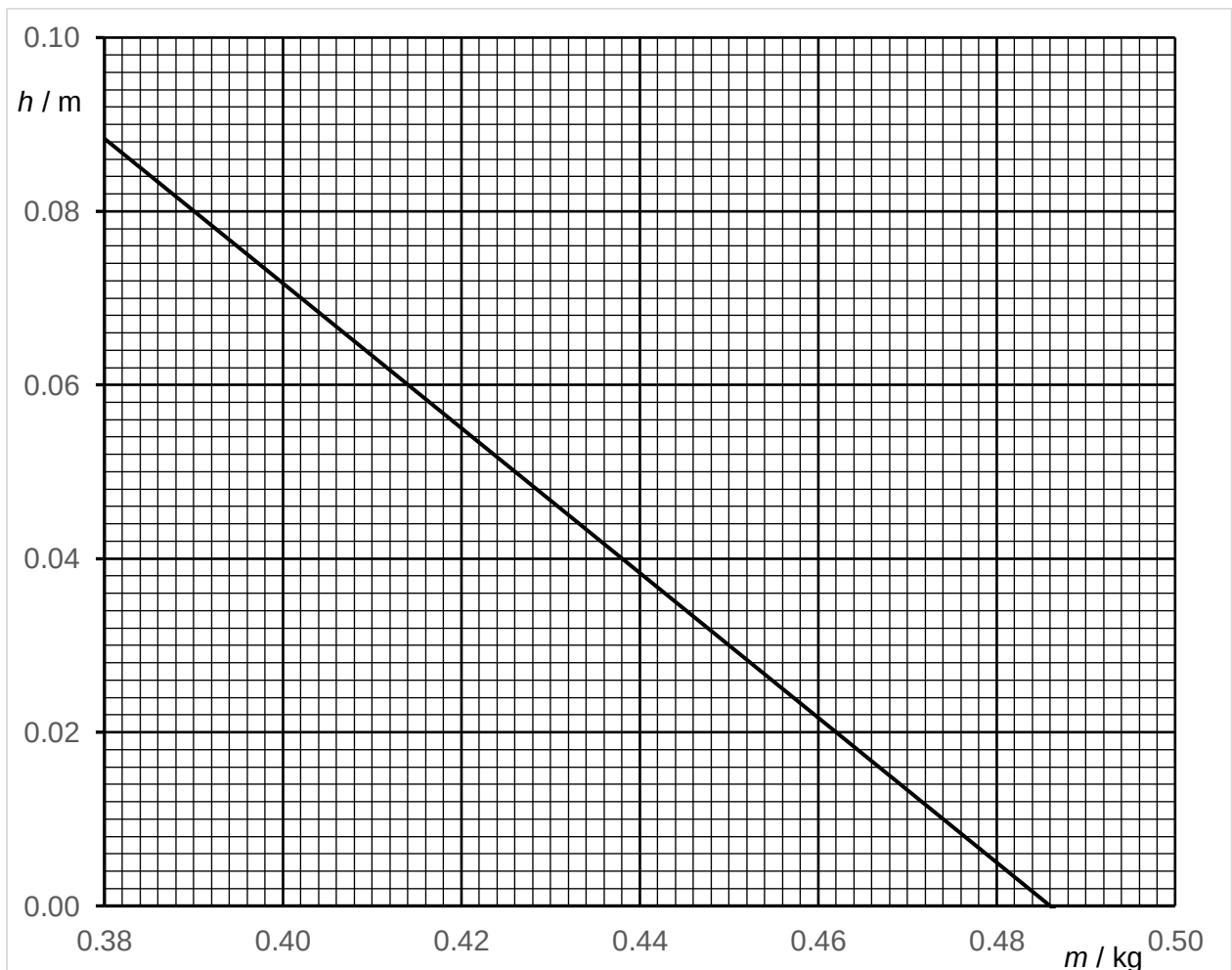


Fig. 2.2

- (a) Using Fig. 2.2, determine the total mass m required to remove the metal block from the water.

From the graph, when h is 0 cm, m is 0.486 kg.

$m = \dots\dots\dots$ kg [1]

- (b) By considering the forces acting on the metal block, show that the gradient of the graph in Fig. 2.2 is represented by $-\left(\frac{1}{1.0 \times 10^3 A}\right)$.

Considering the forces acting on the metal block,

Upthrust + Tension = Weight of block

$$U + T = Mg \quad (\text{where } M \text{ is the mass of block})$$

$$U + mg = Mg \quad (\text{since } T = mg \text{ by considering the forces acting on the mass hanger})$$

$$U = -mg + Mg$$

$$\rho_{\text{water}} Ahg = -mg + Mg \quad (\text{since } U = \text{weight of fluid displaced by Archimedes' Principle})$$

$$\rho_{\text{water}} Ah = -m + M$$

$$h = -\left(\frac{1}{\rho_{\text{water}} A}\right)m + \left(\frac{1}{\rho_{\text{water}} A}\right)M$$

$$\text{Therefore, gradient of graph} = -\left(\frac{1}{\rho_{\text{water}} A}\right) = -\left(\frac{1}{1.0 \times 10^3 A}\right) \quad [\text{shown}]$$

[2]

- (c) Hence, determine the cross-sectional area, A .

From graph,

$$\text{Gradient} = \frac{0.080 - 0.010}{0.390 - 0.474} = -0.833$$

$$-\left(\frac{1}{1.0 \times 10^3 A}\right) = -0.833$$

$$A = -\left(\frac{1}{1.0 \times 10^3 (-0.833)}\right) \\ = 1.2 \times 10^{-3} \text{ m}^2$$

$A = \dots\dots\dots$ m² [2]

(d) Determine the density of the metal block, ρ .

From (a),

mass of block, $M = 0.486$

$$\rho Ah = 0.486$$

$$\rho = \frac{0.486}{Ah} = \frac{0.486}{(1.2 \times 10^{-3})(0.15)} = 2.7 \times 10^3 \text{ kg m}^{-3}$$

$$\rho = \dots\dots\dots \text{ kg m}^{-3} [2]$$

3. (a) Explain the difference between accuracy and precision and state the type of error